Thermodynamics & Thermochemistry

Grade 11 • Chemistry • Session 1

Vedantu

JEE Mains Chemistry 2020-2023 analysis

Chapter name	Thermodynamics_and_Thermochemis
Avg. % weight of chapter	3.0%

If weightage >3% then high weightage chapter, if less than 2% than low weightage chapter

Avg. % Weights of topics in the chapter	
	1.3
Bond_Dissociation_Enthalpy	3.9
Concept_of_Heat_and_Work	7.9
Enthalpy_of_a_System	5.3
Enthalpy_of_Reactions/processes	18.4
Entropy_and_Spontaneity	5.3
First_Law_of_Thermodynamics	1.3
Gibb's_Energy_and_Spontaneity	5.3
Gibb's_Free_Energy_Change_in_Reaction	9.2
Heat_Capacity_of_a_System	1.3
Hess_Law_of_Constant_Heat_Summation	6.6
Internal_Energy	2.6
Introduction_to_Thermodynamics	2.6
Mix	2.6
Relationship_Between_ΔH_and_ΔU	17.1
Reversible_and_Irreversible_Work	7.9
Types_of_Processes	1.3

Learning Outcomes

1. Introduction & Basic Terminologies in Thermodynamics
2. Thermodynamic Processes

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Learning Outcomes

1. Introduction & Basic Terminologies in Thermodynamics
2. Thermodynamic Processes

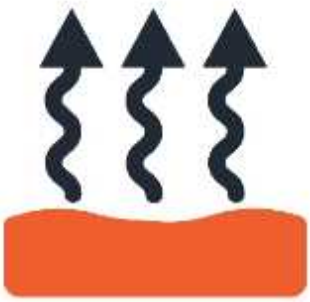
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Thermodynamics

Introduction to Thermodynamics

Therm	+	Dynamics	=	Thermodynamics
"Heat"		"Motion"		Studying movement of heat/energy

HEAT



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Thermodynamics

Introduction to Thermodynamics

- Thermodynamics is a branch of science that deals with energy and its transformation, system and its properties.
- It predicts direction of natural/spontaneous change - like what circumstances are best for **rusting of iron**?



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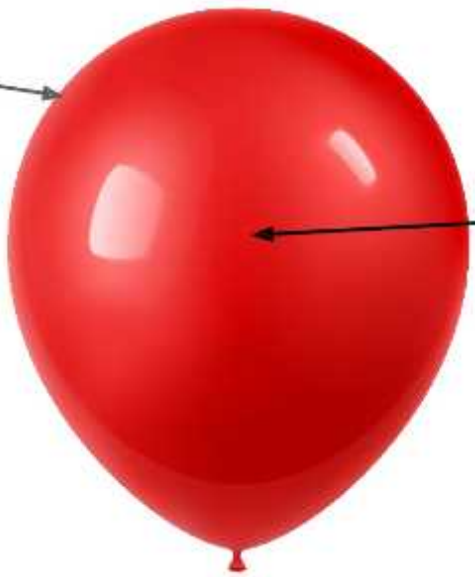
System / Surrounding

Basic Terminologies

System, Surroundings and Boundary

Suppose you want to understand the effect of increase in temperature on air inside a balloon.

Outer surface of balloon will be the **boundary**.



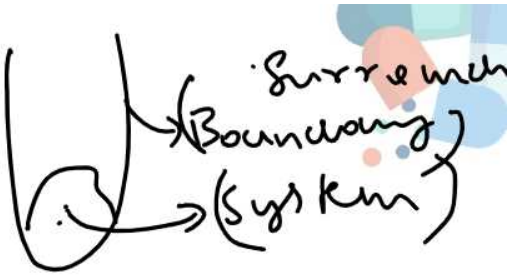
Air inside the balloon will be our **system**.

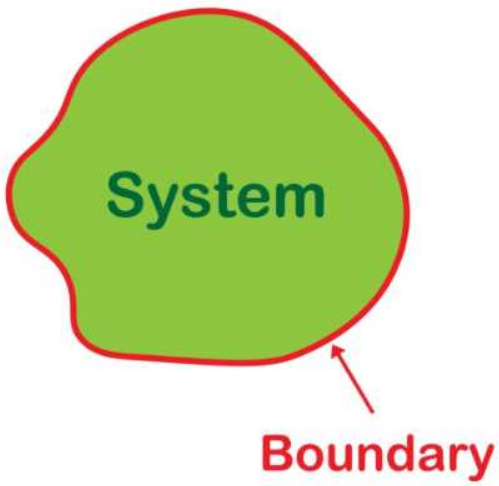
Everything outside the boundary is **surroundings**.

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Basic Terminologies

System, Surroundings and Boundary





- **System** is part of the universe which is arbitrarily set off from the rest of the universe by definite boundaries for the purpose of experimental or theoretical studies.
- The remainder of the universe is the **surroundings** of the system.
- The space separating the system from its surrounding is termed as **boundary**.

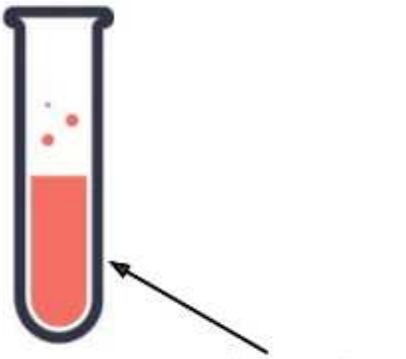
Universe = System + Surrounding

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Basic Terminologies

Types of Boundaries

- Real Boundary
- Imaginary Boundary



Real boundary



Imaginary boundary

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Basic Terminologies

Types of Boundaries

A real boundary is also called as wall. The types of wall are as follows:

			
Rigid wall: The wall (boundary) is immovable.	**Non-Rigid wall:** The wall is movable.	**Adiabatic wall:** The heat can't be exchanged across the wall.	**Diathermic wall:** The heat can be exchanged across the wall.

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Basic Terminologies

Types of System



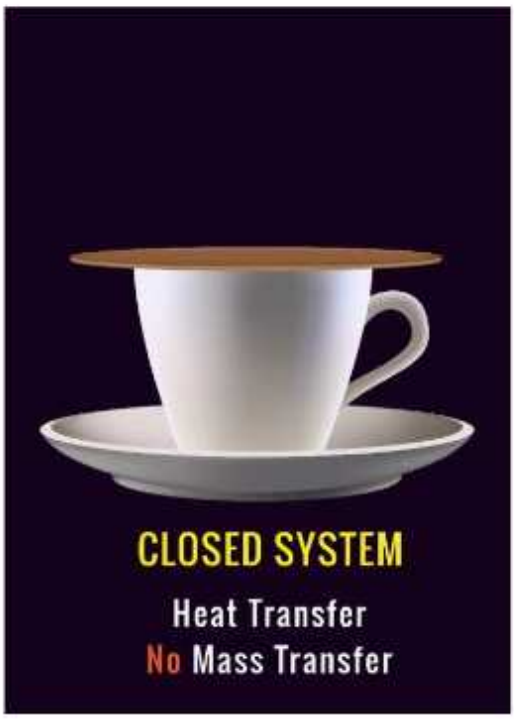
Open System

- A system which can exchange both energy and matter with its surroundings.
- The presence of reactants in an open beaker is an example of an open system.

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Basic Terminologies

Types of System



Closed System

- A system is said to be closed when it permits passage of energy but not mass, across the boundary.
- The presence of reactants in a closed vessel made of conducting material e.g., copper or steel is an example of a closed system.

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Basic Terminologies

Types of System



Isolated System

- A system is said to be isolated when it can neither exchange energy nor matter with its surroundings.
- The presence of reactants in a thermos flask or any other closed insulated vessel is an example of an isolated system.

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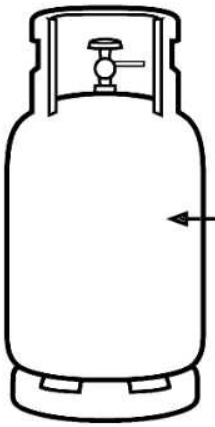
Open System

Basic Terminologies

State of a system

- It means the condition in which the system is present.
- It can be specified/defined by measuring/specifying some observable/measurable properties of the system like pressure, volume, temperature, amount of substance etc.

(T, P, V, Amount of Substance n, mass)



The state of gas inside the cylinder can be defined by mentioning the pressure, volume and temperature.

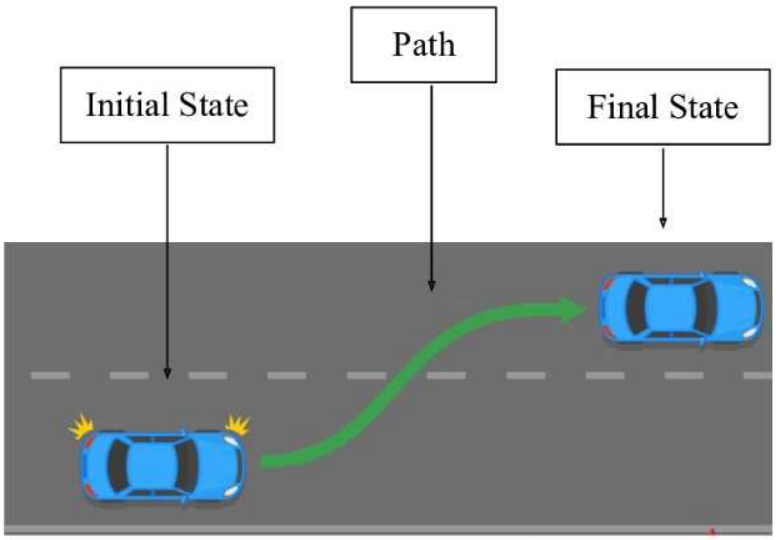
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Basic Terminologies

T_1 to T_2

Path

The series of states passed through by the system during a change from one state to another.



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Basic Terminologies

State Function

Temperature

Pressure

Volume

- Property of a system which is dependent only on the state of the system. It is independent of the path adopted to attain a particular state.
- Changes in state function are not state function. Example: ΔT, ΔV, ΔP etc. are not state functions.

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Basic Terminologies

Path Function

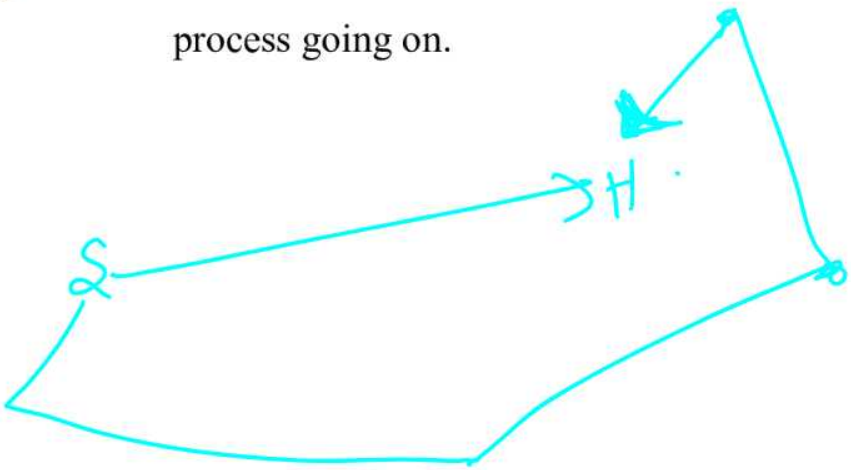


Heat

Work

Distance travelled

- Quantities which are dependent on the path/way the system has achieved a particular state.
- These quantities are defined when there is a process going on.

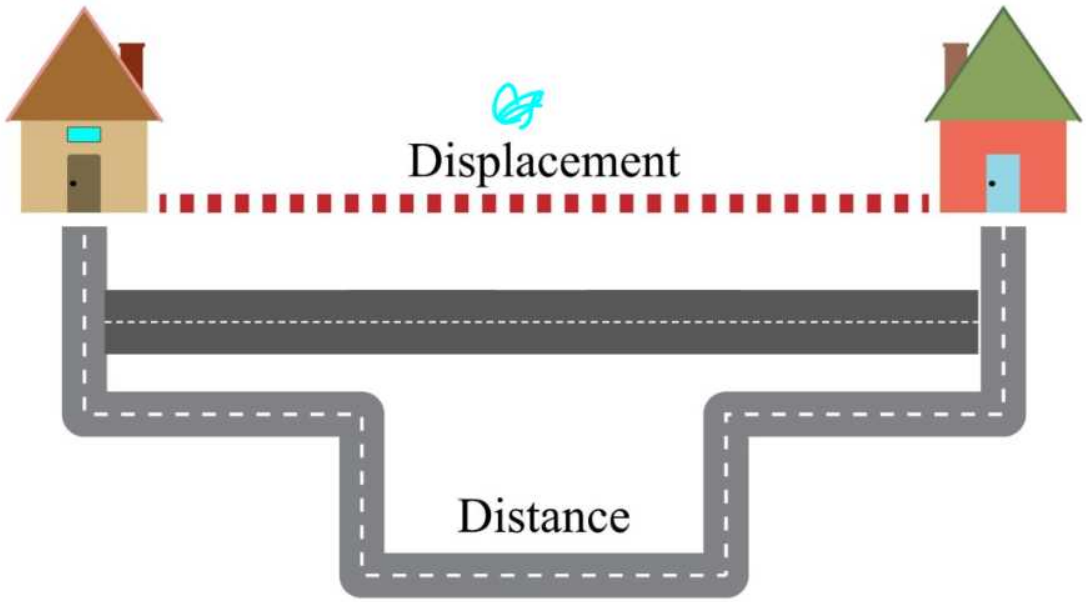


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Basic Terminologies

State and Path Function - Example

Displacement of any object will be a state function but distance travelled by the object will be a path function.

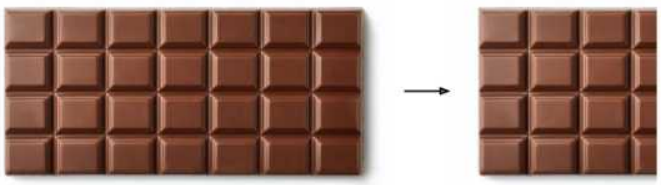


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Basic Terminologies

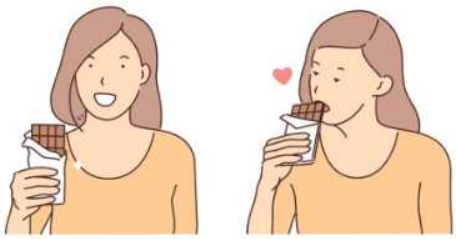
Extensive Properties

- **Depend upon the quantity of the matter contained in the system.**
- The mass of chocolate bar is an example of extensive property because when you eat half of it, the mass becomes half too.



Intensive Properties

- **Independent of the amount of the substance present in the system.**
- The taste of chocolate bar remains the same throughout your eating experience. Thus, it is an intensive property.



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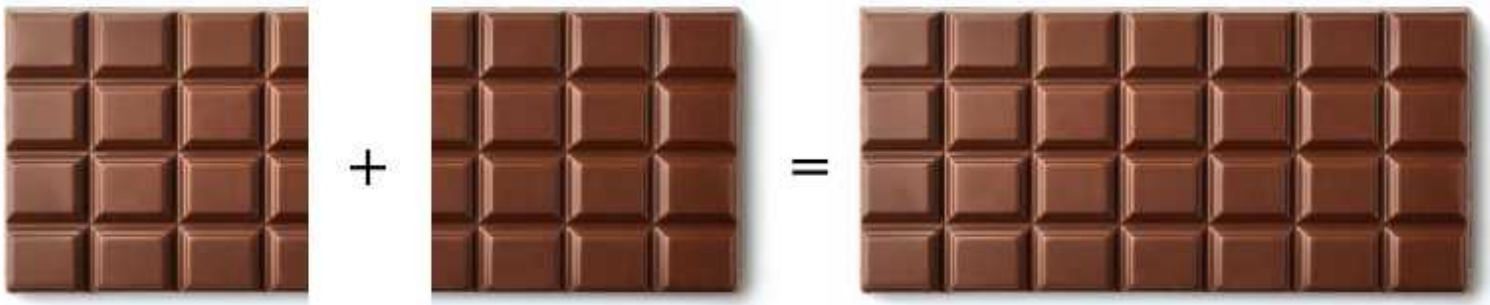
Basic Terminologies

Extensive Properties

- **Additive in nature.**
- On adding two half chocolates to get a full chocolate, the mass of the chocolate becomes two times.

Intensive Properties

- **Non additive in nature.**
- On adding two half chocolates to get a full chocolate, chocolate doesn't get more tasty. The taste remains the same.



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Basic Terminologies

dependant

Extensive Properties

- **Examples** - Volume, mass, total energy etc.

Independant

Intensive Properties

- **Examples** - Temperature, pressure, density, concentration, vapour pressure, B.P., F.P., Viscosity, pH, specific heat etc.

IMPORTANT!

Ratio of two extensive properties is an intensive property.

For example,

Mass	/	Volume	$\frac{\text{Mass}}{\text{Volume}}$ = density
Extensive		**Extensive**	**Intensive**

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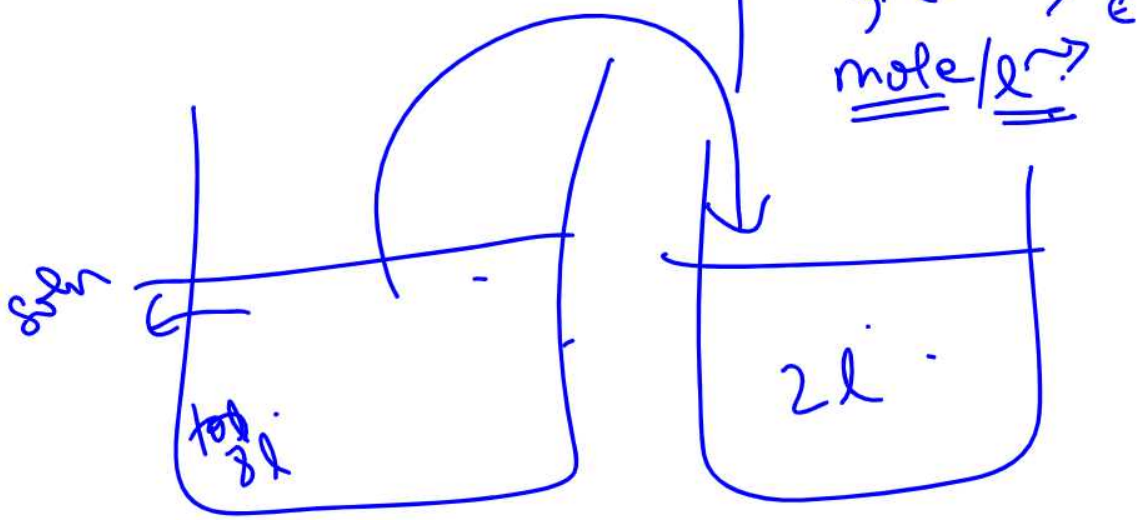
Boiling Point of Substance

a) Intensive

b) Extensive

Why is Concentration intensive property?

mole/l

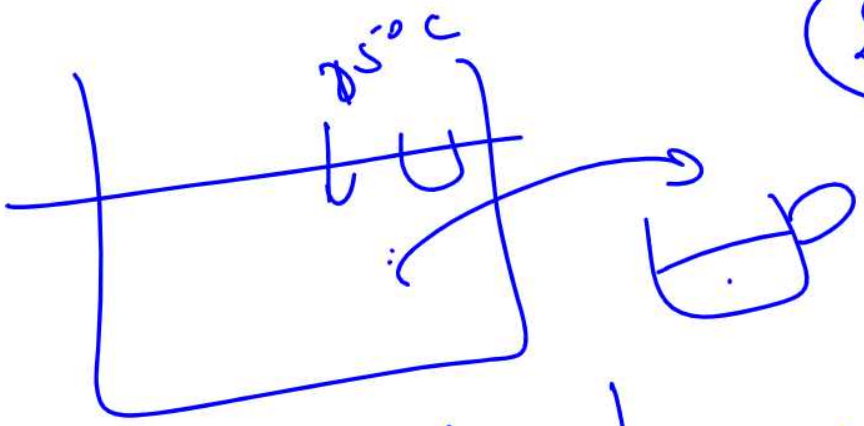


1 g/ml

Temperature

↳ Intensive property.

Density



a) Intensive

b) Extensive

Chat Off

Interactivity Pod

Which thermodynamic system does not exchange energy with its surroundings?

A Closed System

B Isolated System

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Chat On

Interactivity Pod

Which thermodynamic system does not exchange energy with its surroundings?

A Closed System

B Isolated System

An isolated system can neither exchange energy nor matter with its surroundings.

A closed system can exchange energy but not matter with its surroundings.

Let's Solve.



Let's Solve.

Warming ammonium chloride with sodium hydroxide in a test tube is an example of

A. Open system

B. Closed system

C. Isolated system

D. None of these

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Let's Solve.

Warming ammonium chloride with sodium hydroxide in a test tube is an example of

A Open system

B Closed system

C Isolated system

D None of these

Solution

A reaction taking place in a test tube is an example of open system since both mass and heat can be exchanged.

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Doubts Time

I will now be taking frequently asked doubts



Doubts



- How density is intensive, when mass and volume are extensive properties?

- What is the meaning of additive nature of properties?

Quiz.



Quiz.

Chat Off

Which of the following statements is correct?

A. The presence of reacting species in a covered beaker is an example of open system.

B. There is an exchange of energy as well as matter between the system and the surroundings in a closed system

C. The presence of reactants in a closed vessel made up of copper is an example of a closed system.

D. The presence of reactants in a thermos flask or any other closed insulated vessel is an example of a closed system.

Isolated System

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Quiz.

Chat On

Which of the following statements is correct?

A. The presence of reacting species in a covered beaker is an example of open system.

B. There is an exchange of energy as well as matter between the system and the surroundings in a closed system

C. The presence of reactants in a closed vessel made up of copper is an example of a closed system.

D. The presence of reactants in a thermos flask or any other closed insulated vessel is an example of a closed system.

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Quiz.

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Which of the following statements is correct?

Solution

The corrected statements are -

(A) Presence of reacting species in a covered beaker is an example of closed system.

(B) There is an exchange of energy but not of matter between system and the surroundings in a closed system.

(C) The presence of reactants in a closed vessel made of copper is an example of a closed system.

(D) The presence of reactants in a thermos flask or any other closed insulated vessel is an example of an isolated system.

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Quiz.

Chat Off

A thermodynamic state function is

A One which obeys all the law of thermodynamics

B A quantity which is used in measuring thermal changes

C One which is used in thermochemistry

D A quantity whose value depends only on the state of system

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Quiz.

Chat On

A thermodynamic state function is

A. One which obeys all the law of thermodynamics

B. A quantity which is used in measuring thermal changes

C. One which is used in thermochemistry

D. A quantity whose value depends only on the state of system

Solution

State functions are the properties of a system which is dependent only on the state of the system.

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Quiz.

Chat Off

Which is the intensive property?

A Temperature

B Viscosity

C Density

D All of the above

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Quiz.

Chat On

Which is the intensive property?

A Temperature

B Viscosity

C Density

D All of the above

Solution

Temperature, viscosity, density all are independent of the amount of the substance.

Thus, all are intensive properties.

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Milestone Achieved

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Learning Outcomes

1. Introduction & Basic Terminologies in Thermodynamics
2. **Thermodynamic Processes**

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Let's Discuss

Give two examples each of extensive and intensive properties.

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Thermodynamic Processes

The operation by which a system changes from one state to another state is called a process.

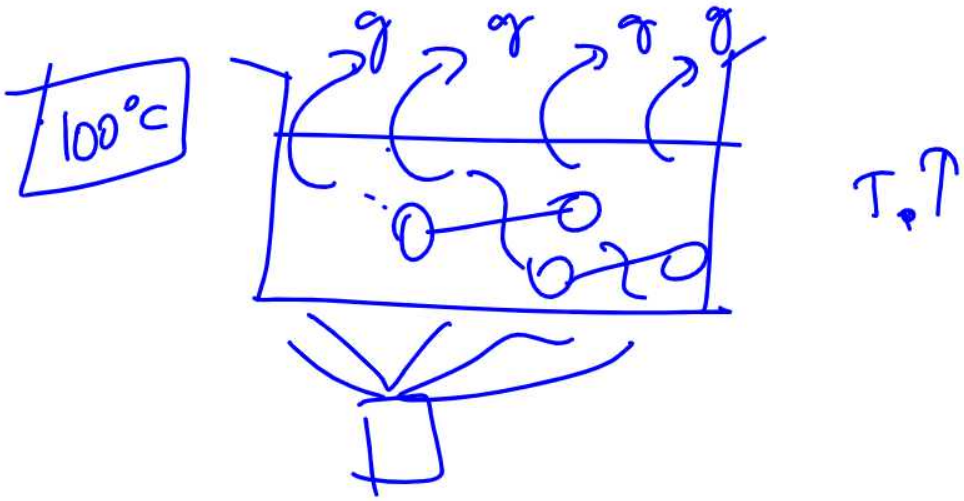
Isothermal Process:
dT = 0
T = constant

Isobaric Process:
dP = 0
P = constant

Isochoric Process:
dV = 0
V = constant

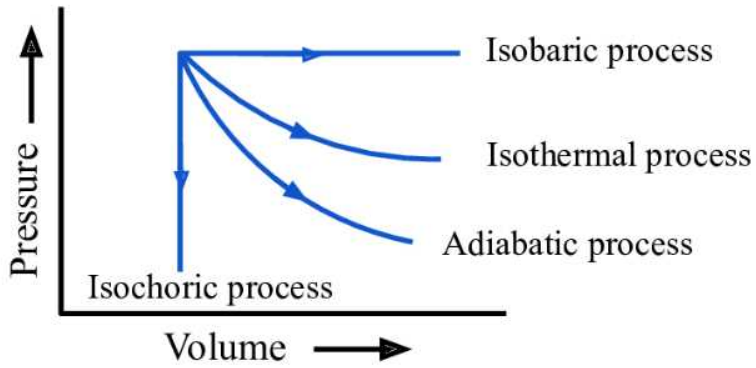
Adiabatic Process:
dQ = 0
Q = 0

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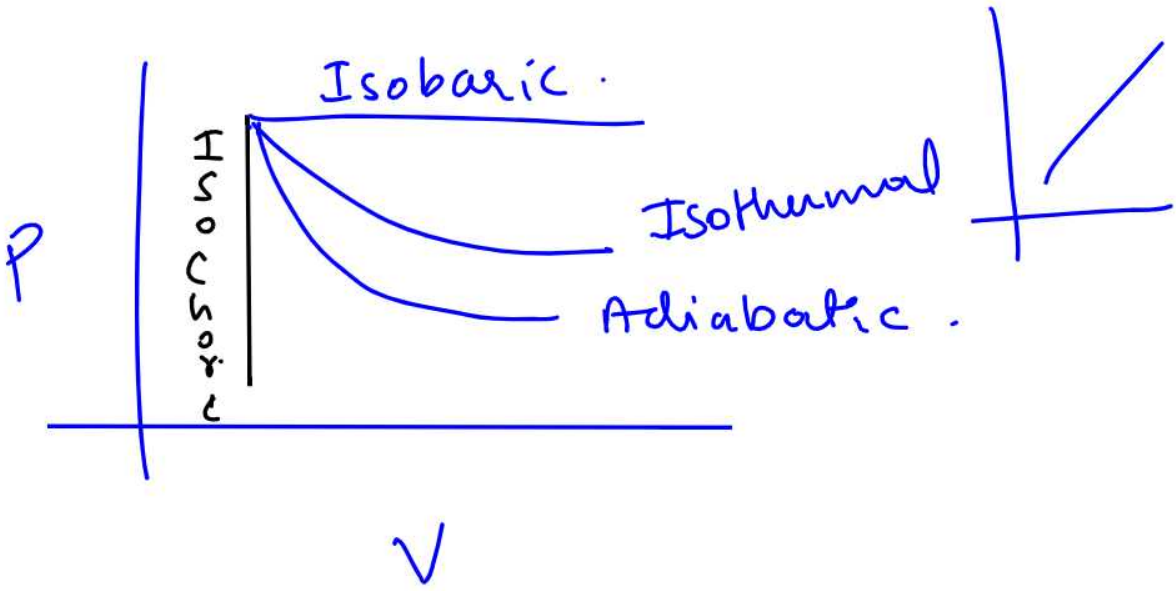

100°C
q
q
q
q
T.P

Thermodynamic Processes

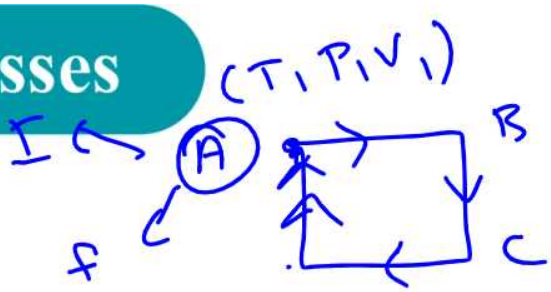
Graphical Representation



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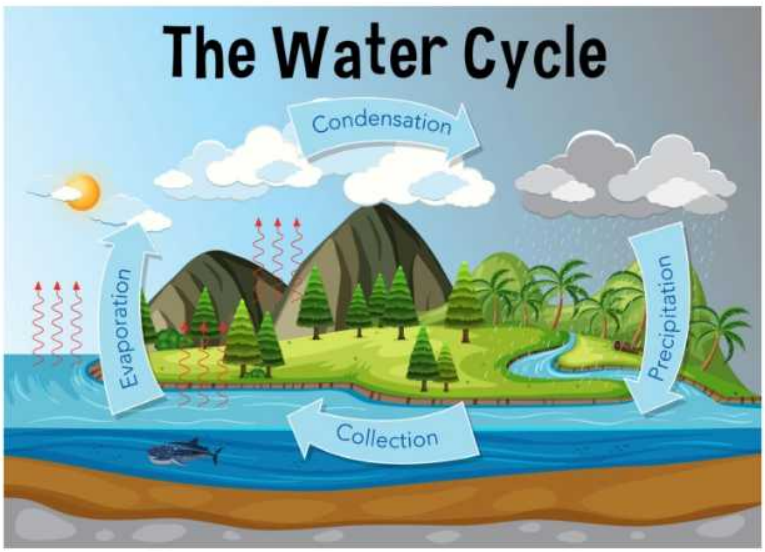

Isobaric.
Isochoric
Isothermal
Adiabatic.
P
V

Thermodynamic Processes



Cyclic Process

- When a system undergoes a number of different states by different processes and finally returns to its initial state, it is said to have completed a cycle and the process is called a cyclic process.
- For a cyclic process, the change in state function is zero, i.e., $\Delta T = 0$, $\Delta V = 0$, $\Delta P = 0$.



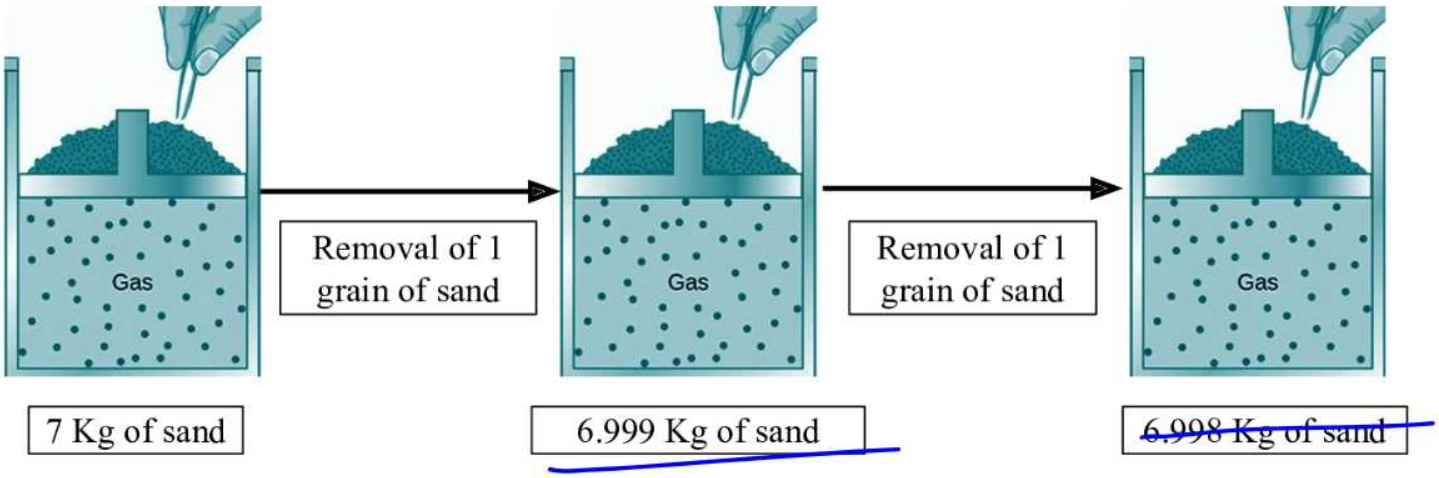
$\Delta T = 0$ $\Delta V = 0$ $\Delta P = 0$

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Thermodynamic Processes

Reversible process

To get an idea of a reversible process we can consider the following system -



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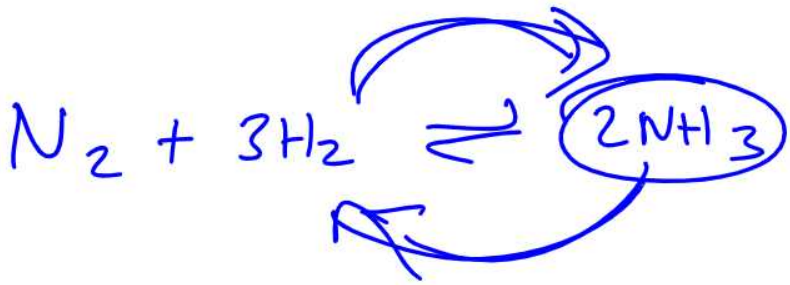
Thermodynamic Processes

Features of reversible process

The process that can be reversed by a very small change is known as reversible process.

- If a process is carried out in such a manner so that the system is always in equilibrium (driving force = opposing force) at every stage of the process, it is reversible.
- Process takes place at infinitesimally slow rate.
- An ideal reversible process will take infinite time to get completed.
- There is no ideal reversible process in universe.

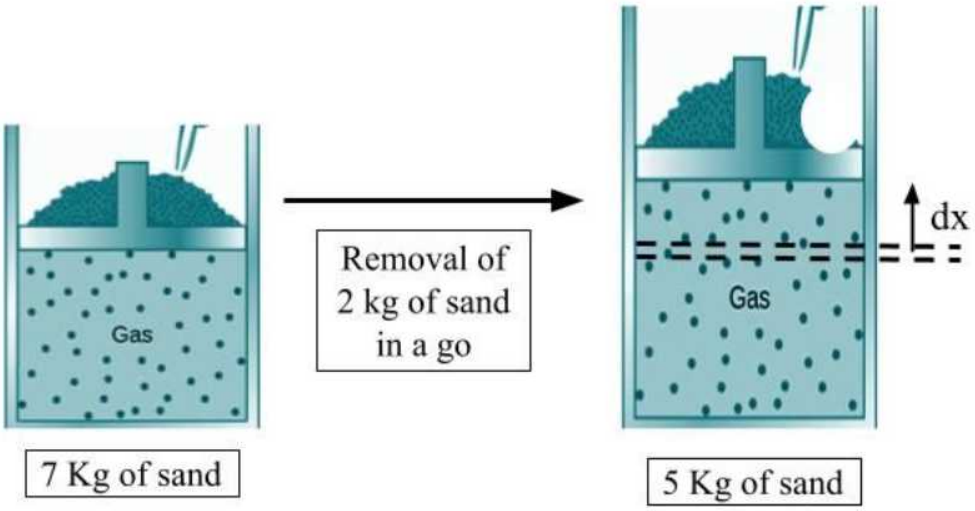
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$N_2 + 3H_2 \rightleftharpoons 2NH_3$

Thermodynamic Processes

Irreversible process

To get an idea of irreversible process we can consider the following system -



Thermodynamic Processes

Features of irreversible process

The process which cannot be reversed by a small change is known as irreversible.

- Irreversible processes will get completed in finite time.
- All real process are irreversible.
- If expansion/compression takes place against constant external pressure then it is irreversible.
- If there is sudden change then the process is irreversible.

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Chat Off

Interactivity Pod

Temperature remains constant during an adiabatic process.

A True

B False

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Chat On

Interactivity Pod

Temperature remains constant during an adiabatic process.

A True

B False

Temperature remains constant during an isothermal process.

In an adiabatic process, there is no exchange of heat between the system and its surroundings.

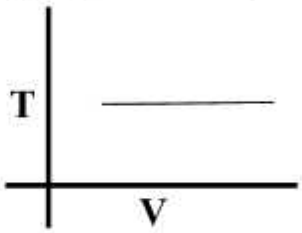
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Let's Solve.



Let's Solve.

The given graph corresponds to which equation?



A. $V = 0$

B. $PV = \text{constant}$

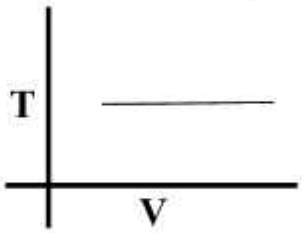
C. $V/T = \text{constant}$

D. $PT = \text{constant}$

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Let's Solve.

The given graph corresponds to which equation?



A. $V = 0$

B. $PV = \text{constant}$

C. $V/T = \text{constant}$

D. $PT = \text{constant}$

Solution

In the given graph temperature remains constant with variation in volume. So the process is isothermal and $PV = \text{constant}$.

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Silent Mode

Chat On

Isothermal process = Temperature constant

Adiabatic process = Pressure constant

(True/False)



Discuss among yourselves

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Doubts Time

I will now be taking frequently asked doubts



Doubts



- How cyclic process is different from reversible process ?
- How expansion/compression taking place against constant external pressure is irreversible ?

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Quiz.



Quiz.

Chat Off

$dq = 0$

When no heat energy is allowed to enter or leave the system, it is called

A Isothermal process

B Reversible process

C Adiabatic process

D Irreversible process

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Quiz.

Chat On

When no heat energy is allowed to enter or leave the system, it is called

A. Isothermal process

B. Reversible process

C. Adiabatic process

D. Irreversible process

Solution

For adiabatic processes, heat change(q) = 0.

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Quiz.

Chat Off

$V \propto T$ (P constant).

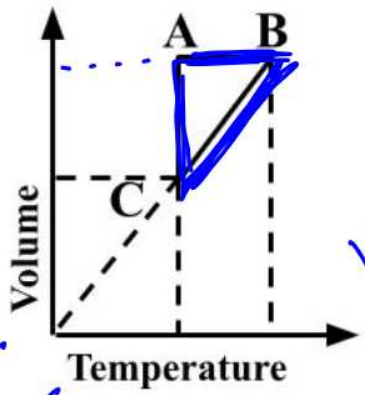
Five moles of a gas is put through a series of changes as shown graphically in a cyclic process. The processes A → B, B → C and C → A respectively are

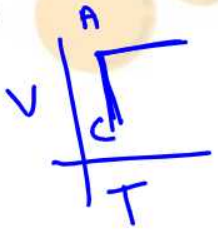
A. Isochoric, Isobaric, Isothermal

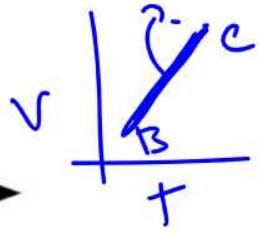
B. Isobaric, Isochoric, Isothermal

C. Isothermal, Isobaric, Isochoric

D. Isochoric, Isothermal, Isobaric







A → B (Isochoric)

A → C (Isothermal)

B → C (Isobaric)

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Quiz.

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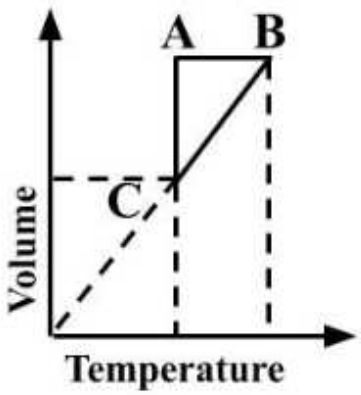
Five moles of a gas is put through a series of changes as shown graphically in a cyclic process. The processes A → B, B → C and C → A respectively are

A. Isochoric, Isobaric, Isothermal

B. Isobaric, Isochoric, Isothermal

C. Isothermal, Isobaric, Isochoric

D. Isochoric, Isothermal, Isobaric



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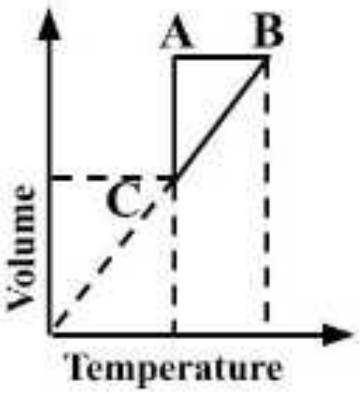
Quiz.

Chat On

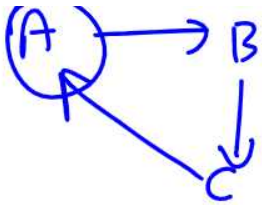
Five moles of a gas is put through a series of changes as shown graphically in a cyclic process. The processes A → B, B → C and C → A respectively are

Solution

For process A → B, volume is constant. Thus, it is isochoric.
For process B → C, pressure is constant ($V \propto T$). Thus, it is isobaric.
For process C → A, temperature is constant. Thus, it is isothermal.



Quiz.



Chat Off

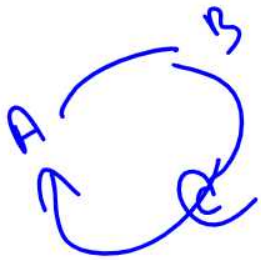
A gaseous system changes from state A (P_1, V_1, T_1) to B (P_2, V_2, T_2), B to C (P_3, V_3, T_3) and finally from C to A. The whole process may be called.

A. Reversible process

B. Cyclic process

C. Isobaric process

D. Spontaneous process



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Quiz.

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A gaseous system changes from state A (P_1, V_1, T_1) to B (P_2, V_2, T_2), B to C (P_3, V_3, T_3) and finally from C to A. The whole process may be called.

A. Reversible process

B. Cyclic process

C. Isobaric process

D. Spontaneous process

Solution

Since the initial and final state of the system is the same, it is a cyclic process.

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Milestone Achieved

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Summary.



Summary



Rest is surrounding

Universe = system + surrounding

Open System: Exchanges heat and mass	**Types of System**
Closed System: Exchanges heat but not mass	
Isolated System: Neither heat nor mass exchanged	

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Summary

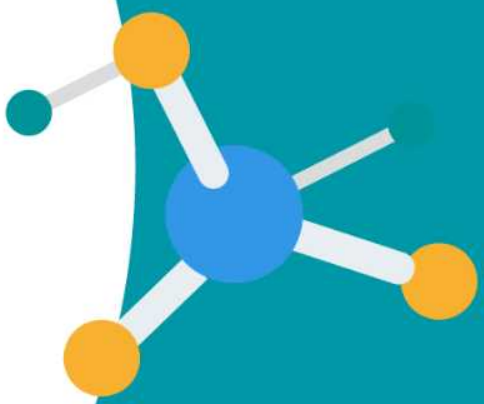
Types of thermodynamic variables	State Function
	Path function

Types of thermodynamic properties	Extensive Property
	Intensive Property

Types of processes	
	Isobaric Process ($\Delta P = 0$)
	Isothermal Process ($\Delta T = 0$)
	Isochoric Process ($\Delta V = 0$)
	Adiabatic Process ($q = 0$)
	Reversible Process
	Irreversible Process
	Cyclic Process ($\Delta T = 0$, $\Delta U = 0$, $\Delta P = 0$)

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Summary.



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Summary

Zeroth Law of Thermodynamics

If two systems are in thermal equilibrium with third system, then they are also in thermal equilibrium with each other.

First Law of Thermodynamics

Energy can neither be created nor be destroyed.

$\Delta U = q + w$

Quantity	Positive Sign (+)	Negative Sign (–)
For q	System gains heat	System loses heat
w	Work done on system	Work done by system
ΔU	Net gain of energy by system	Net loss of energy by system

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HOMEWORK

(Pick and Solve Based on Session Coverage)

Topic name as per Curriculum tree	Assignment Detail	Tatva Exercise 1 question number
Introduction to thermodynamics	Assignment 1 (question 1, 3, 4, 12)	1, 2, 3, 4, 5, 6,
Types of processes	Assignment 1 (question 5, 6, 7, 8) Assignment 4 (question 13, 16, 17)	
Internal energy	Assignment 1 (question 9, 10, 13, 16) Assignment 2 (question 3, 9, 12)	7, 8, 9
Concept of Heat and work	Assignment 1 (question 2, 11, 14, 15) Assignment 2 (question 10)	10, 11, 12, 13
Zeroth law of thermodynamics	Assignment 2 (question 1)	
First law of thermodynamics	Assignment 2 (question 2, 7, 11, 13, 14)	14, 15, 16, 17
Reversible and irreversible work	Assignment 2 (question 4, 5, 6, 8, 15) Assignment 4 (question 18, 19)	18, 19, 20, 21, 22, 23, 24, 25, 26, 27
Heat Capacity of A System	Assignment 3 (question 1, 2, 3, 4, 12, 13) Assignment 4 (question 14)	28, 29
Adiabatic process and poisson's ratio	Assignment 3 (question 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 17)	30, 31, 32, 33, 34
Enthalpy of A System	Assignment 4 (question 6, 7, 8, 11, 12)	35, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48,
Gibb's Free Energy Change in Reaction	Assignment 6 (question 7, 14)	

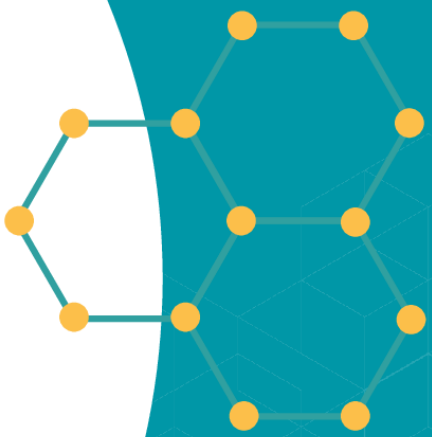
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Feedback



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Thanks.



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